A recurrence whose power ranks detect perfect squares

Principia Math in Progress

28 September 2026

Task 1: resolve MathDB 366795

The rank of a constant-recursive sequence is the order of its minimal eventual constant-coefficient linear recurrence. Conjecture 5.3 of Rowland and Sistos Barrón [1] predicts eventual polynomiality for general rank sequences and eventual pseudo-polynomiality for particular rank sequences, which allow coefficient cancellations. We disprove the latter assertion, and hence the combined conjecture.

Here pseudo-polynomiality is understood in the periodic-coefficient, or quasipolynomial, sense: there are a positive integer p and polynomials $P_0, \dots, P_{p-1}$ such that the sequence equals $P_a(M)$ for all sufficiently large $M \equiv a \pmod p$. This makes explicit the interpretation suggested by the periodic cancellation examples immediately preceding the conjecture. The source does not separately define the term.

Theorem 1. *The integer constant-recursive sequence*

$$s(n) = 1 + 2^n + 3^n - 6^n + n27^n + n54^n \qquad (n \geq 0) \tag{1}$$

has rank eight. For every integer $M \geq 5$, the rank $R(M)$ of the sequence $(s(n)^M)_{n\geq 0}$ is

$$R(M) = \frac{(M+1)(3M^2+3M+2)}{2} - 3(M-1)\mathbf{1}_{2|M} - 6(M-2)\mathbf{1}_{M \text{ is a square}}. \tag{2}$$

In particular, R is not eventually quasipolynomial.

Rank from exponential polynomials

We use the elementary fact that, for distinct nonzero numbers $\lambda_1, \dots, \lambda_v$ and nonzero polynomials A_j,

$$\operatorname{rank}\left(\sum_{j=1}^{v} A_j(n)\lambda_j^n\right) = \sum_{j=1}^{v}(\deg A_j + 1). \tag{3}$$

For completeness, let E be the shift operator. On $A(n)\lambda^n$, $E - \lambda$ lowers the polynomial degree by one until it annihilates the term; $E - \mu$, with $\mu \neq \lambda$, preserves its degree and nonzero leading coefficient. Applying products of these operators isolates each distinct exponential root and also proves their linear independence. Consequently an annihilating polynomial in E must contain $(E - \lambda_j)^{\deg A_j + 1}$ for each j, and their product suffices. The same argument applies to eventual recurrences.

Thus (1) has minimal characteristic polynomial

$$(z-1)(z-2)(z-3)(z-6)(z-27)^2(z-54)^2,$$

which has integer coefficients and a nonzero constant term. All terms of s are integers, so it meets the source's hypotheses.

Three leading coefficient types

Introduce independent indeterminates t, x, y and put

$$F(t,x,y) = (1+x) + y(1-x) + ty^3(1+x).$$

Then $s(n) = F(n, 2^n, 3^n)$. Write

$$F(t,x,y)^M = \sum_{i,j} C_{i,j}(t) x^i y^j.$$

Unique prime factorization makes all numbers $2^i 3^j$ distinct. Therefore (3) gives

$$R(M) = \sum_{C_{i,j} \neq 0} (\deg C_{i,j} + 1). \tag{4}$$

Choosing h factors of the third summand, k of the second, and $M-h-k$ of the first gives the exact expansion

$$F^M = \sum_{\substack{h,k \geq 0 \\ h+k \leq M}} \frac{M!}{h!k!(M-h-k)!}\, t^h y^{3h+k} (1+x)^{M-k} (1-x)^k. \tag{5}$$

Write a prospective y-exponent uniquely as $j = 3h + r$, with $r \in \{0,1,2\}$. Its largest possible t-exponent is h. It is feasible exactly when $h + r \leq M$: using $h - d$ third summands instead would require $r + 3d$ second summands and a total of $h + r + 2d$ factors. Hence the feasible pairs are precisely

$$r \in \{0,1,2\}, \qquad 0 \leq h \leq M - r. \tag{6}$$

For each pair and each $0 \leq i \leq M$, the top coefficient of $C_{i,3h+r}(t)$ is a positive multinomial multiple of

$$A_r(i) := [x^i](1+x)^{M-r}(1-x)^r.$$

Define the auxiliary polynomials

$$K_k(i) := [z^k](1-z)^i(1+z)^{M-i}, \qquad a = M - 2i.$$

Expansion of the binomial coefficients shows

$$\binom{M}{r} A_r(i) = \binom{M}{i} K_r(i). \tag{7}$$

Indeed, matching the summand with b negative choices on each side gives the same factorial expression. In particular,

$$K_0 = 1, \qquad K_1 = a, \qquad K_2 = \frac{a^2 - M}{2}. \tag{8}$$

Thus the leading coefficient never vanishes for $r = 0$; for $r = 1$ it vanishes only at $i = M/2$, when M is even; and for $r = 2$ it vanishes only at

$$i = \frac{M - \sqrt{M}}{2}, \quad \frac{M + \sqrt{M}}{2},$$

when M is a square. For square $M \geq 5$ these are two distinct integers in $[0, M]$, since M and $\sqrt{M}$ have the same parity.

Exact losses, including the boundary

At a vanishing leading coefficient, a term of degree $h-1$ is available exactly when

$$h \geq 1 \quad \text{and} \quad h \leq M-r-2. \tag{9}$$

By (5) and (7), it is a nonzero constant multiple of $K_{r+3}(i)$ unless that polynomial vanishes. Differentiating the defining generating polynomial gives

$$(k+1)K_{k+1} = aK_k - (M-k+1)K_{k-1}.$$

Consequently,

$$K_4(i) = \frac{M(M-2)}{8} \qquad \text{when } a=0, \tag{10}$$

$$K_5(i) = \frac{a(M-1)(M-4)}{20} \qquad \text{when } a^2 = M. \tag{11}$$

For example, at $a^2 = M$ the recurrence successively gives $K_3 = -(M-1)a/3$ and $K_4 = -M(M-1)/12$, followed by (11). The quantities in (10)–(11) are nonzero in the applicable cases for $M \geq 5$. Thus under (9), a vanished leading term reduces the rank contribution from $h+1$ to h, an exact loss of one.

When $h = 0$, a vanished constant coefficient also loses one rank unit. At the final two feasible values $h = M-r-1, M-r$, no smaller t-exponent is possible at all: replacing h by $h-d$, $d \geq 1$, would require $h+r+2d > M$ factors. The whole polynomial is then zero, so its losses are respectively $M-r$ and $M-r+1$. For each vanishing index i, the aggregate loss over all feasible h is therefore

$$\underbrace{(M-r-1)}_{h=0,\ldots,M-r-2} + (M-r) + (M-r+1) = 3(M-r). \tag{12}$$

There is one such index for $r=1$ at even M, and two for $r=2$ at square M. These are disjoint y-exponent classes, so their losses add even when M is an even square.

If no leading terms vanished, (4) would equal

$$\begin{aligned}(M+1)\sum_{r=0}^{2}\sum_{h=0}^{M-r}(h+1) &= \frac{M+1}{2}\big((M+1)(M+2) + M(M+1) + (M-1)M\big) \\ &= \frac{(M+1)(3M^2+3M+2)}{2}.\end{aligned}$$

Subtracting exactly the losses in (12) proves (2).

Why no eventual quasipolynomial can agree

If R were eventually quasipolynomial, subtracting its polynomial and parity terms would make

$$D(M) = 6(M-2)\mathbf{1}_{M \text{ is a square}}$$

eventually quasipolynomial, with some positive period p. On the residue class 0 (mod p), its constituent polynomial would vanish at all sufficiently large integers $(pt)^2 + p$: these lie strictly between $(pt)^2$ and $(pt+1)^2$, and hence are nonsquares. That polynomial would have infinitely many roots, so would be zero. But on the same residue class it must take the nonzero value $6((pt)^2 - 2)$ at all sufficiently large squares $(pt)^2$. This contradiction completes Theorem 1.

Scope and verification. All 23 pages of the current manuscript were read, and the separate 23-page author copy was fully reconciled. No journal version was identified. An existing MathDB submission under problem 366796 claims the general, no-cancellation case and expressly leaves the particular case open; our argument addresses that remaining case. The accompanying verifier expands the six-term polynomial directly, checks ranks with exact integer coefficients, and verifies the auxiliary identities symbolically. These finite checks support the all-M proof above. Source and claim-search limits appear in the separate audit. This is author-agent review, not independent peer review.

Tasks 2–5: proposed follow-up problems

These are separate research proposals, not completed results.

2. Determine the minimum rank of an integer constant-recursive sequence whose power-rank sequence is not eventually quasipolynomial.

3. Decide whether every integer constant-recursive sequence with only simple characteristic roots has an eventually quasipolynomial power-rank sequence.

4. Determine the power-rank sequence of $1 + 2^n + 3^n - 6^n$ by classifying the integer zeros in the corresponding Krawtchouk coefficient arrays.

5. For each fixed $d \geq 3$, construct an integer constant-recursive sequence with an explicit rank formula containing a nonzero correction supported exactly on the perfect dth powers, or prove an obstruction to such a construction.

References

[1] Eric Rowland and Jesús Sistos Barrón, *Complexity of powers of a constant-recursive sequence*, arXiv:2501.14643v1 (2025), Conjecture 5.3, p. 15.